Field	Details
Date	19 July 2026
Team ID	SB-AIML-2026-0042
Project Name	Emotion Detector - AI-Driven Emotion Detection & Personalized Learning Support P
Maximum Marks	3 Marks

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1. Project Description

Emotion Detector is an AI-powered web platform that analyzes a student's free-text description of a study problem, detects their emotional state using a dual deep learning ensemble (BiLSTM + BERT) enhanced by rule-based keyword matching, and generates personalized learning guidance using the Google Gemini API.

The platform supports five emotional states relevant to learning:

- Bored** - Student finds the material unengaging
- Confident** - Student has a clear understanding
- Confused** - Student lacks clarity on a concept
- Curious** - Student is engaged and wants to explore
- Frustrated** - Student is stuck and emotionally distressed

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2. Problem Statement

Students in digital learning environments struggle silently. They cannot identify or communicate their emotional state effectively, and existing platforms deliver generic content regardless of the learner's affect. This leads to reduced learning outcomes, burnout, and disengagement.

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3. Proposed Solution

An AI platform that:

- Accepts free-text description of a study problem
- Detects emotional state (5 classes) with confidence scores
- Applies rule-based keyword enhancement for accuracy
- Detects mixed emotion states
- Generates structured, empathetic, personalized guidance via Gemini API
- Logs all interactions for analytics and trend visualization

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4. Technology Stack

Category	Technology
Language	Python 3.10+
ML Models	BiLSTM (TensorFlow/Keras), BERT (HuggingFace/PyTorch)
Generative AI	Google Gemini API (gemini-pro)
Web App	Streamlit
Visualization	Plotly
Dataset	dair-ai/emotion (HuggingFace, 16K samples)
Deployment	Streamlit Community Cloud (Free)

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5. System Architecture

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User Input (Text)


Preprocessing (NLTK)

     BiLSTM Inference  P1
     BERT Inference    P2


Ensemble (40%  P1 + 60%  P2)


Rule-Based Keyword Boost


Mixed Emotion Detection


Final Emotion + Confidence

     Gemini API  5-Section Guidance


Streamlit UI Display + CSV Log
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6. Key Features

S.No	Feature	Status
1	Free-text emotion input	Implemented
2	BiLSTM emotion classification	Implemented
3	BERT emotion classification	Implemented

S.No	Feature	Status
4	Weighted ensemble (40/60)	Implemented
5	Rule-based keyword boost	Implemented
6	Mixed emotion detection	Implemented
7	Gemini personalized guidance	Implemented
8	Interactive Streamlit UI	Implemented
9	Analytics dashboard	Implemented
10	Streamlit Cloud deployment	Implemented

7. Results

Metric	Value
Validation Accuracy (Ensemble)	88.1%
Test Accuracy (Ensemble)	87.6%
Macro F1-Score	0.87
End-to-End Response Time	~4.2 seconds
Total Features Implemented	20/20 (100%)

8. Deployment

Field	Value
Platform	Streamlit Community Cloud
GitHub Repo	https://github.com/Harsh-maker007/Smartbridge_project1
API Used	Google Gemini API (Free Tier)
Cost	Zero - entirely free infrastructure

9. Conclusion

The Emotion Detector successfully addresses the gap in emotion-aware personalized learning. By combining classical NLP, deep learning, and generative AI, it delivers a user-friendly, accessible, and intelligent platform that adapts its support to the student's current emotional state - available 24/7 via a public web URL at zero cost.